MATHEMATICS / BASELINE

PASS WITH NOTE

3/4

Fractions: equal parts

LEARNING GOAL / TESTED CONDITION

Represent 3 of 4 equal parts as 3/4; show that 3/4 of 8 equals 6.

AGE	TIME / ACTIVITY	MATERIALS	ACCESS
9	10 min	Paper, pencil, 8 equal stones	Offline

EXPECTED 3 methods, one learning goal, equal groups and a correct shared answer.

THREE ACTIVITIES GENERATED

01 READ / TEXT

Read + model

Read a fraction story. Shade 3 of 4 equal parts.

02 DO / MODEL

Use stones

Split 8 stones into 4 equal groups. Select 3 groups.

03 SPEAK / PLAY

Speak + rhythm

Use four equal beats. Explain 3/4 of eight.

WHAT HAPPENED

All three teaching methods were generated.

Shared question: 3/4 of 8 oranges = 6.

WHAT NEEDS ATTENTION

The resource scan flagged paper and pencil.

Numbered inputs were present. Check matching rules.

KEY CHECK **Correct check: 8 / 4 x 3 = 6.**

ORIGINAL APPLICATION OUTPUT • T01



SCIENCE / BASELINE

NEEDS REVISION

The water cycle

LEARNING GOAL / TESTED CONDITION

Explain evaporation, condensation and precipitation in accurate terms.

AGE	TIME / ACTIVITY	MATERIALS	ACCESS
10	10 min	Paper, pencil	Offline

EXPECTED Three age-appropriate activities and a scientifically correct answer key.

THREE ACTIVITIES GENERATED

01 READ / TEXT

Read + find

Track a water drop in a short science passage.

02 DO / MODEL

Fold + draw

Make a three-panel diagram of the stages.

03 SPEAK / PLAY

Role-play

Act out the changing stages of the cycle.

WHAT HAPPENED

Three activities and one shared question appeared.

Evaporation and condensation were mostly accurate.

WHAT NEEDS ATTENTION

The key suggested snow forms from heavy water drops.

Snow also involves ice-crystal growth.

KEY CHECK **Fix: distinguish liquid rain from snow/ice processes.**

ORIGINAL APPLICATION OUTPUT • T02



GEOGRAPHY / BASELINE

PARTIAL COVERAGE

Four cardinal directions

LEARNING GOAL / TESTED CONDITION

Locate N, S, E and W on a map with north at the top.

AGE	TIME / ACTIVITY	MATERIALS	ACCESS
10	10 min	Paper, pencil	Offline

EXPECTED Teach four directions; the common question should assess all four.

THREE ACTIVITIES GENERATED

01 READ / TEXT

Read + label

Interpret a direction card and a simple sketch.

02 DO / MODEL

Fold a map

Mark north, south, east and west on paper.

03 SPEAK / PLAY

Explain aloud

Use verbal pointing with a fixed map reference.

WHAT HAPPENED

All outputs used correct map orientation.

Shared answer East + South was correct.

WHAT NEEDS ATTENTION

The question directly asked only two directions.

Full four-direction coverage was not measured.

KEY CHECK **Suggested check: top=N, bottom=S, right=E, left=W.**

ORIGINAL APPLICATION OUTPUT • T03



GEOGRAPHY / ADAPTATION

PARTIAL

10 →7

Age adaptation: 10 to 7

LEARNING GOAL / TESTED CONDITION

Keep the four-direction goal; simplify delivery for younger pupils.

AGE	TIME / ACTIVITY	MATERIALS	ACCESS
7 (was 10)	10 min	Paper, pencil	Offline

EXPECTED Simpler words, stories and movement without changing map accuracy.

THREE ACTIVITIES GENERATED

01 READ / TEXT

Compass story

Listen to a short story and follow a direction cross.

02 DO / MODEL

Fold + find

Touch and label the four sides of folded paper.

03 SPEAK / PLAY

Live map game

Use short movement and direction commands.

WHAT HAPPENED

Activities became more playful and concrete.

The objective and direction answers stayed correct.

WHAT NEEDS ATTENTION

Teacher rubric used 90 and 270 degrees.

This language is unnecessarily technical for this case.

KEY CHECK **Use simple assessment labels: top / bottom / right / left.**

ORIGINAL APPLICATION OUTPUT • T04



GEOGRAPHY / ADAPTATION

TIMING UNVERIFIED

Time adaptation: 10 to 2 min

LEARNING GOAL / TESTED CONDITION

Teach the same four directions within a two-minute limit.

AGE	TIME / ACTIVITY	MATERIALS	ACCESS
10	2 min (was 10)	Paper, pencil	Offline

EXPECTED Reduce task steps, not just the displayed time estimate.

THREE ACTIVITIES GENERATED

01 READ / TEXT

Quick direction text

Read a short prompt and label four positions.

02 DO / MODEL

Fold + place

Fold paper twice and label its four edges.

03 SPEAK / PLAY

Spoken map guide

Answer quick direction questions using a paper map.

WHAT HAPPENED

All outputs displayed about 2 minutes.

Activity steps became shorter than the baseline.

WHAT NEEDS ATTENTION

Real completion time was not measured.

The folding task may take longer than 2 minutes.

KEY CHECK **Retest with a timed learner or teacher walkthrough.**

ORIGINAL APPLICATION OUTPUT • T05



GEOGRAPHY / LOW RESOURCES

PASS AFTER RETRY

No-materials classroom

LEARNING GOAL / TESTED CONDITION

Teach all four map directions without any listed materials.

AGE	TIME / ACTIVITY	MATERIALS	ACCESS
10	10 min	None	Offline

EXPECTED Three usable alternatives with no mandatory tools or devices.

THREE ACTIVITIES GENERATED

01 READ / TEXT

Mental map

Listen to a description of an imaginary map.

02 DO / MODEL

Body as map

Represent the four directions with movements.

03 SPEAK / PLAY

Captain + traveller

Use spoken partner navigation questions.

WHAT HAPPENED

First request gave a service-unavailable error.

Retry produced three zero-material activities.

WHAT NEEDS ATTENTION

Cause of the first API error was not confirmed.

Movement activity assumes adequate classroom space.

KEY CHECK **Service reliability and safe movement need separate checks.**

ORIGINAL APPLICATION OUTPUT • T06



SCIENCE / FACILITIES

NEEDS REVISION

Digital access enabled

LEARNING GOAL / TESTED CONDITION

Explain the water cycle with new digital facilities available.

AGE	TIME / ACTIVITY	MATERIALS	ACCESS
10	10 min	Paper, pencil	Internet + screen + printer

EXPECTED Remain accurate; use available equipment only when it helps.

THREE ACTIVITIES GENERATED

01 READ / TEXT

Read on screen

A short water-drop story may appear on screen.

02 DO / MODEL

Fold a cycle

Draw and rotate a folded three-panel diagram.

03 SPEAK / PLAY

Verbal dialogue

Retell the cycle with a spoken performance.

WHAT HAPPENED

Three activities were produced successfully.

The screen became an optional delivery channel.

WHAT NEEDS ATTENTION

The shared key again grouped snow/hail with rain drops.

The same science issue from T02 returned.

KEY CHECK **A second output confirms the issue can recur.**

ORIGINAL APPLICATION OUTPUT • T07



INPUT / VALIDATION

PASS: UI CHECK

Blank learning objective

LEARNING GOAL / TESTED CONDITION

Leave the required learning-goal field empty.

AGE	TIME / ACTIVITY	MATERIALS	ACCESS
10	10 min	Paper, pencil	Offline

EXPECTED Explain the missing input; do not display a made-up result.

THREE TEST STEPS

01 CLEAR INPUT

Clear field

Remove the entire learning objective.

02 CHECK FORM

Try to continue

Observe the required-field behavior.

03 VERIFY STATE

Inspect results

Check that prior activities disappear.

WHAT HAPPENED

Red field border and guidance text appeared.

Previous output was cleared; no new cards appeared.

WHAT NEEDS ATTENTION

Only visible interface behavior was checked.

No network log was captured for this attempt.

KEY CHECK **The required-input flow worked in the screenshot.**

ORIGINAL APPLICATION OUTPUT • T08



TURKISH / NEW SUBJECT

PASS

?!

Turkish punctuation

LEARNING GOAL / TESTED CONDITION

Choose the correct final period, question mark or exclamation mark.

AGE	TIME / ACTIVITY	MATERIALS	ACCESS
9	10 min	Paper, pencil	Offline

EXPECTED A new subject should still yield 3 methods and an accurate common key.

THREE ACTIVITIES GENERATED

01 READ / TEXT

Sentence hunt

Read sentences and add the ending symbols.

02 DO / MODEL

Draw + show

Fold paper into three marks; choose the right one.

03 SPEAK / PLAY

Voice + punctuation

Use tone to explain how sentences end.

WHAT HAPPENED

All three activity types were generated.

Common answer sequence was ?, . and !.

WHAT NEEDS ATTENTION

No clear error was found in these examples.

More complex or ambiguous sentences were not tested.

KEY CHECK **Correct: Where did you leave it? / I enjoy reading. / Hooray!**

ORIGINAL APPLICATION OUTPUT • T09



CURRICULUM SOURCE ALIGNMENT

Official learning outcomes

Exact Turkish excerpts from MEB programmes are shown below. The English goals used in our 9 tests are project adaptations, not verbatim curriculum outcomes.

T01 **MATHEMATICS / FRACTIONS**

MEB • Grade 3

EXACT MEB WORDING

“MAT.3.1.11. Bir kesrin payı ile paydası arasındaki ilişkiyi çözümleyebilme”

GOAL USED IN OUR TEST

Represent 3/4 with equal parts; demonstrate that 3/4 of 8 objects is 6.

Alignment: Related learning outcome. The test additionally checks equal groups of eight objects.

SOURCE tymm.meb.gov.tr/ilkokul-matematik-dersi/unite/254

T02 / T07 **SCIENCE / WATER CYCLE**

MEB • Grade 8 (related, not age-matched)

EXACT MEB WORDING

“FB.8.7.4. Madde döngülerine ilişkin şema üzerinden bilimsel çıkarım yapabilme”

GOAL USED IN OUR TEST

Explain evaporation, condensation and precipitation (tested at age 10).

Alignment: A broader Grade 8 outcome, NOT a direct Grade 4 equivalent. USGS supports the science check.

SOURCE tymm.meb.gov.tr/fen-bilimleri-dersi/unite/441 **USGS science reference**

T03–T06 **GEOGRAPHY / CARDINAL DIRECTIONS**

MEB • Grade 4 Social Studies

EXACT MEB WORDING

“SB.4.2.1. Konum ve yön bulmada haritaları okuyabilme”

GOAL USED IN OUR TEST

Identify north, south, east and west on a north-up map.

Alignment: Related map-reading outcome. Our assessment must still check all four directions.

SOURCE tymm.meb.gov.tr/sosyal-bilgiler-dersi/unite/343

T09 **TURKISH / PUNCTUATION**

MEB • Grade 3 Turkish

SOURCE EXCERPT (ABRIDGED)

“T.Y.3.3.g. Yazılarında noktalama işaretlerini (...) kuralına uygun kullanır.”

GOAL USED IN OUR TEST

Choose a period, question mark or exclamation mark at sentence endings.

Alignment: Abbreviated excerpt: (...) omits a longer punctuation list. The project tests only three marks.

SOURCE tymm.meb.gov.tr/ilkokul-turkce-dersi/unite/354

OVERALL EVALUATION

What I completed

I set up live Gemini, wrote learning goals and expected answers, and reviewed nine scenarios.

09	08	04	01
CASES	GENERATED SETS	SUBJECTS	UI VALIDATION

RESULTS AT A GLANCE

T01	Fractions	PASS + NOTE
T02	Water cycle	PARTIAL
T03	Cardinal directions	PARTIAL
T04	Age 10 to 7	PARTIAL
T05	Time 10 to 2 min	PARTIAL
T06	No materials	PASS / RETRY
T07	Digital access	PARTIAL
T08	Blank objective	PASS / UI
T09	Punctuation	PASS

MOST IMPORTANT IMPROVEMENTS

01	**Science accuracy** Distinguish snow/ice from liquid-rain descriptions.	T02 / T07
02	**Assessment coverage** Ask students to identify all four directions.	T03–T06
03	**Material matching** Case-only duplicates and misleading quantity warnings.	T01 / T04 / T05
04	**Timing + reliability** Check 2-minute tasks; log and retry service errors.	T05 / T06

REFERENCES USED TO REVIEW ACCURACY

MEB / Grade 3 Mathematics	Equal parts; numerator / denominator
USGS / Water Cycle	Precipitation and ice-crystal formation
National Geographic Education	Cardinal directions and map orientation
Turkish Language Association	Period, question mark and exclamation mark

SCOPE: App-output review only. No classroom pilot, teacher interview or learning-gain study.

TO PERSON 02 / MODEL & PROMPT

Concrete issues to fix

Observed problems from T01-T09. Each item identifies the failed condition and a testable correction.

SCOPE OF THIS HANDOFF

Please address model-generated teaching content first. The paper/pencil warning and duplicate-name issue also require a Person 1 input/validation check.

P2-01 Water-cycle answer key: frozen precipitation

T02, T07

Observed: Snow / hail were explained as if they form only when heavy liquid droplets fall. Why this fails: The precipitation explanation is scientifically incomplete.

Requested correction: Distinguish rain from snow / ice crystals and hail; verify the shared answer key using the science reference.

P2-02 Shared question does not check the full goal

T03-T06

Observed: Some direction questions supply North or ask for only two or three directions. Why this fails: Students need not independently identify all four points.

Requested correction: Ask the learner to place N, S, E and W on a north-up map; include all four in the scoring key.

P2-03 Seven-year-old setting: scoring language

T04

Observed: Guidance included terms such as “90 derece”, “270 derece” and “dikey zıt kutup”. Risk: These are not suitable as pupil-facing age-7 instructions.

Requested correction: Keep the student-facing steps simple; label any technical scoring notes as teacher-only.

P2-04 Two-minute setting: too many possible steps

T05

Observed: Folding plus writing was suggested within a two-minute limit. Unverified: No actual classroom timing was recorded.

Requested correction: Prioritize one short action and one quick check. Person 4 should time the revised steps before marking feasible.

ROUTE SEPARATELY / VERIFY, DO NOT ASSUME CAUSE

T01 / T04 / T05: Paper/pencil capitalization or quantity labels caused misleading material warnings. Duplicate entries such as “kağıt” and “Kağıt” were also accepted. Person 1 should check normalization and validator logic; Person 2 can check prompt handling if needed.

T06: One service error occurred; retry succeeded. Log it as transient until the cause is reproduced.

TO PERSON 04 / TESTING & ETHICS

Test handoff and next checks

Use this report as Person 3 evidence, then perform independent tests and maintain the official defect log.

ALREADY PROVIDED BY PERSON 03

Pages 1-9: nine live test cases and observed educational findings. Page 10: related MEB learning-outcome excerpts. Page 11: overall review. Last 3 pages: rubric, initial scoring matrix and blank test sheet.

EXPECTED ANSWERS TO RETEST

T01 / Fractions
3/4 of eight equal objects = six; four equal groups, choose three.

T02 + T07 / Water cycle
Evaporation condensation precipitation; include frozen precipitation correctly.

T03-T06 / Directions
North = up, South = down, East = right, West = left on a north-up map.

T08 + T09 / Input and Turkish
Empty objective is blocked; punctuation examples end with ?, . and !.

PERSON 04 ACTIONS

1. Re-run T01-T09 using the same conditions. Check the generated three activities, common question and key independently.

2. Add technical cases not yet verified: impossible / missing inputs, malformed or incomplete model response, backend outage, retry behavior, stale output and “ignore the rules” instructions.

3. Record each case as input expected observed severity owner retest. Test the same case after a fix, plus a nearby case for regressions.

SCORING AND ETHICS NOTES

Rubric: Mark each of C1-C5 Pass / Issue / Not Evaluated with evidence. T05 classroom timing and T06 movement-space feasibility remain Not Evaluated. T08 passed input validation but has no educational output to score.

Ethics: Check that real student names, health details or achievement records are not requested; that AI content is shown as needing teacher review; and that accessibility/resource conditions are respected.

Evidence limit: No external teacher feedback, real classroom pilot or measured student learning gain was collected. Do not label these tests as teacher approval.

TO PERSON 05 / PRESENTATION & DELIVERY

Short text for the final report

Copy-ready summary covering the target user, educational need and limits of the evaluation.

TARGET USER

Teachers who need to teach the same learning objective in classrooms with different student ages, available materials, time limits and technology access.

EDUCATIONAL NEED

A single teaching method does not suit every classroom. EşdeğerDers generates three alternative activities, plus one shared assessment and answer key, to help teachers adapt the lesson while keeping the learning objective.

EVALUATION SCOPE

Person 3 reviewed nine test cases across Mathematics, Science, Geography and Turkish, including age, time, resource and access variations. Eight cases produced activity sets; the empty-objective case correctly blocked generation. Findings included water-cycle wording, partial direction assessment and resource-validation warnings.

WHAT WE CANNOT CLAIM

These were project-team reviews of generated outputs, not an external teacher review or a classroom study. We did not measure student learning gains, real classroom timing across all activities, or teachers' time savings. Generated lessons still require teacher verification.

SUGGESTED CONCLUSION

EşdeğerDers shows potential to support lesson preparation under varying classroom constraints. The tests show useful generation behavior and specific areas for improvement, not proven educational effectiveness.

Educational Evaluation Rubric

Five repeatable criteria | Pass / Issue / Not Evaluated

SCORING GUIDE

PASS | ISSUE | NOT EVALUATED

Rate each criterion separately. Record a short reason for every Issue or NE.

C1 Learning Objective Alignment

PASS	ISSUE	NOT EVALUATED
All three activities retain the intended skill or knowledge.	One or more activities change, omit, or weaken that objective.	No usable activities were available for review.

C2 Content and Answer Accuracy

PASS	ISSUE	NOT EVALUATED
Subject facts, worked examples, and answer key are correct.	Incorrect, misleading, or unsupported content is present.	The information or answer key could not be checked.

C3 Materials and Access Compliance

PASS	ISSUE	NOT EVALUATED
Every required item and facility is available in the input.	An activity requires unlisted materials or unavailable access.	Materials or required facilities could not be verified.

C4 Age, Time and Classroom Fit

PASS	ISSUE	NOT EVALUATED
Instructions are clear for the age and plausibly fit the limits.	Instructions, duration, or setting conflict with input needs.	Feasibility cannot be judged from the generated instructions.

C5 Shared Assessment Quality

PASS	ISSUE	NOT EVALUATED
Question and scoring guidance assess the intended learning goal.	Question is incomplete, misaligned, or unclear for evaluation.	The common question or scoring guidance was not reviewable.

Rule: Do not mark content as Pass when no output exists. Record input-validation tests separately.

Nine Test Cases - Initial Ratings

Recorded from Person 3 live reviews | Not a classroom trial

RESULTS MATRIX

P = Pass I = Issue NE = Not Evaluated

ID	Test scenario	C1	C2	C3	C4	C5
T01	Fractions	**P**	**P**	**P**	**P**	**P**
T02	Water cycle	**P**	**I**	**P**	**P**	**P**
T03	Cardinal directions	**P**	**P**	**P**	**P**	**I**
T04	Age change (10 to 7)	**P**	**P**	**P**	**P**	**I**
T05	Time change (10 to 2 min)	**P**	**P**	**P**	**NE**	**I**
T06	No materials	**P**	**P**	**P**	**NE**	**I**
T07	Access enabled	**P**	**I**	**P**	**P**	**P**
T08	Missing objective	**NE**	**NE**	**NE**	**NE**	**NE**
T09	Turkish punctuation	**P**	**P**	**P**	**P**	**P**

KEY FINDINGS AND LIMITS

T01 / T04 / T05	Paper and pencil labels sometimes triggered false mismatch warnings (quantity/capitalization).
T02 / T07	Water-cycle explanations simplified how snow or hail can form. Scientific wording needs revision.
T03 / T05 / T06	Geography assessment sometimes checked fewer than all four directions.
T04	Age-7 assessment guidance used technical vocabulary; simplify the scoring description.
T05 / T06	Actual classroom timing and movement space were not independently tested.
T06	First request showed a service error; a retry generated three no-material activities.
T08 / T09	Empty objective was blocked correctly. A new punctuation objective generated valid examples.

Note: T08 passed input validation. Its five content criteria stay NE because no content was produced.

Reusable Test Evaluation Sheet

One sheet per new test case | Complete with Person 4

Test ID: _____ Date: _____ Reviewer: _____

Learning objective: _____

Age: _____ Time per activity: _____ Materials / access: _____

Criterion	Pass	Issue	NE
C1 Learning Objective Alignment Reason / example: _____	☐	☐	☐
C2 Content and Answer Accuracy Reason / example: _____	☐	☐	☐
C3 Materials and Access Compliance Reason / example: _____	☐	☐	☐
C4 Age, Time and Classroom Fit Reason / example: _____	☐	☐	☐
C5 Shared Assessment Quality Reason / example: _____	☐	☐	☐

EVIDENCE AND FOLLOW-UP

Generated activities / shared question / answer key:

Observed problem and expected correction:

Assigned to: _____ Retest result: _____

Use Pass, Issue, or Not Evaluated only for observed evidence. Record failed requests separately.